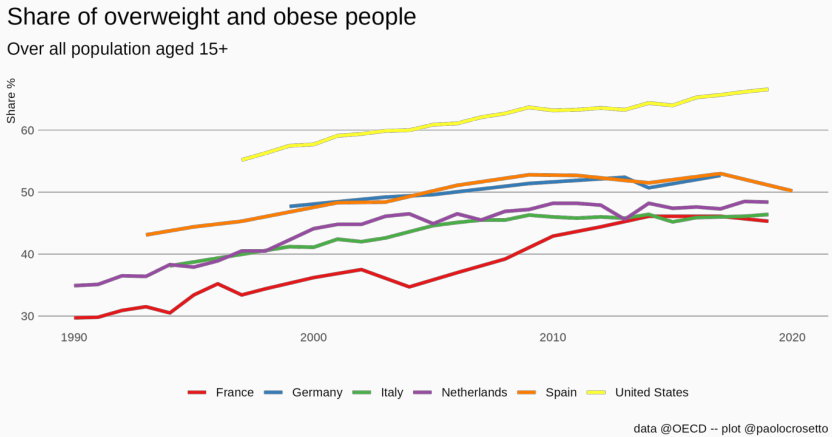
Using the laboratory to investigate food choice and labeling

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ESRI, Dublin, 25 October 2022

Intro: the problem

An obesity explosion



WHO: medical cost of obesity in the U.S.: $147-210 bln

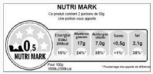
Policy tools

- Regulation
- Information
- Labeling
- Price policies
- Nudges
- ...

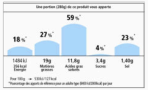
This talk

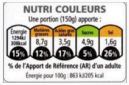
- Regulation
- Information
- **Labeling**
- **Price policies**
- Nudges
- …











\+

Fat **tax** & thin **subsidy**

A series of key policy questions

- Do **labels** work?
- Which label design is the **best** to **impact** choice?
- **How much** of an impact labels have?
- Do **price** interventions work?
- Do they work **better** or **worse** than labels?
- How do the two policies **interact**?

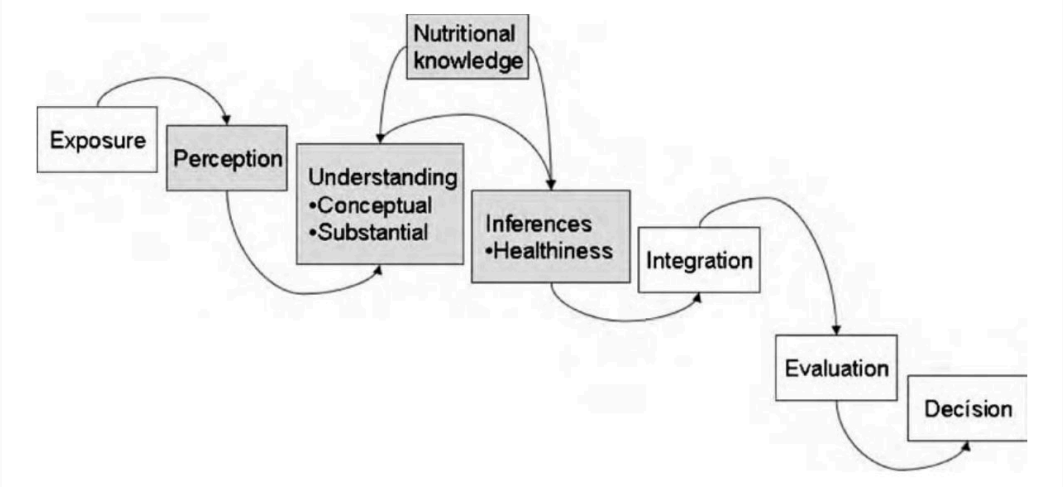
Intro: which role for the lab?

What can we learn from the lab?

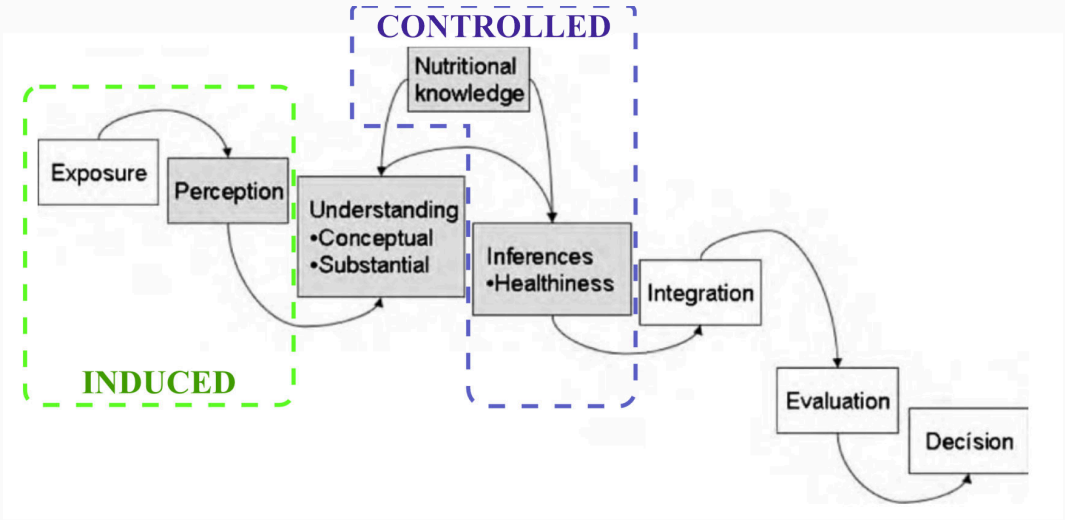




A conceptual framework (Grunert)



A conceptual framework (Grunert)



Two main roles of the lab in an applied policy context

Getting into the mind of subjects

- focus on cognitive aspects
- clearly identify mechanisms
- (if needed) sidestep preferences
- heuristics, choice processes



Two main roles of the lab in an applied policy context

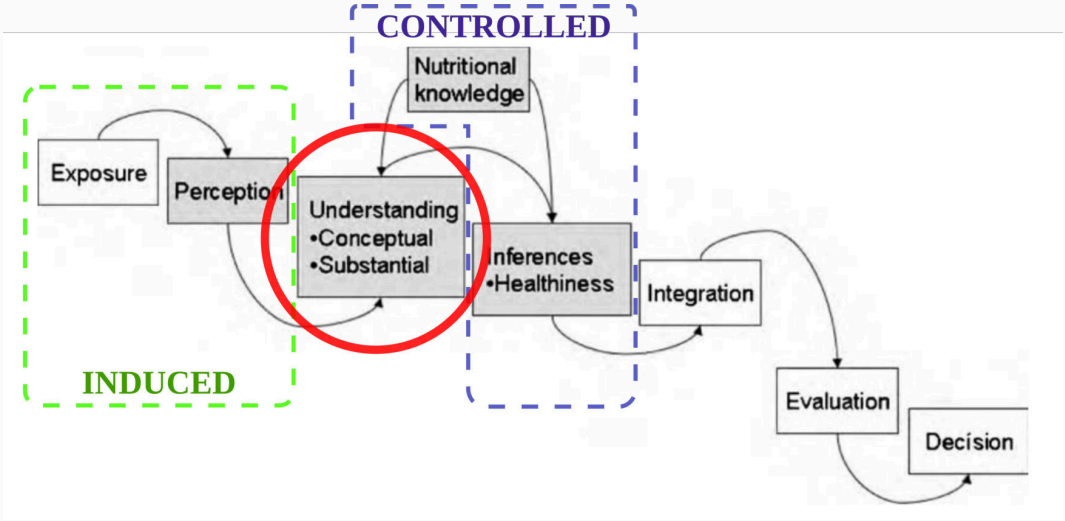
Building counterfactuals

- explore different scenarios
- integrate preferences with control
- track macro consequences
- cheaply explore solutions



Part 1: getting into the mind of subjects

Getting into the mind of subjects



The usual design



The usual design



Study 1: building *diets*
[JoEP 2015 – L. Muller, B. Ruffieux]

What label is better to build a healthy diet?

Diets

- a diet is a complex object, akin to a portfolio
- you won't die for one bad item, but if the overall balance is wrong

Task

- subject "hired as a nutritionist for a canteen"
- must compose daily menu
- menu must satisfy nutritional constraints
- subject guided by labels: **numbers**, **colors**, or **both**.

Incentives

- If the daily diet built satisfies nutritional constraints $\Rightarrow$ flat fee (2 euro)
- Several daily diets to build

Our design: diet-building



Characteristics:

- no preferences
- incentivized
- "realistic"

We add:

- labels
- constraints

Dimensions

Nutrition is multidimensional. We consider three cases:

1-dimension Kcal only are displayed.

4-dimension Kcal + 'bad' nutrients: salt, sugar, fat.

7-dimension 4d + 'good' nutrients: vitamin C, fiber, calcium.

Labels can have numbers, or colors, or both:

Numbers modeled on Guideline Daily Amounts / Reference Intakes

Colors modeled on Traffic Lights

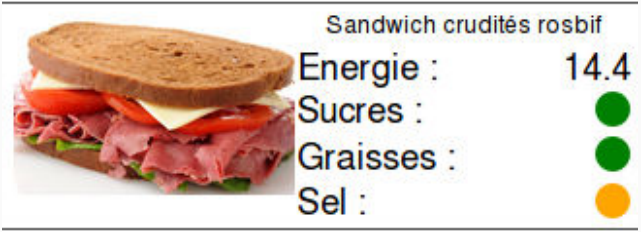
Num+col both of the above combined

Numbers



Tarte aux poireaux	
Energie :	12.3
Sucres :	1.7
Graisses :	46.4
Sel :	19.1

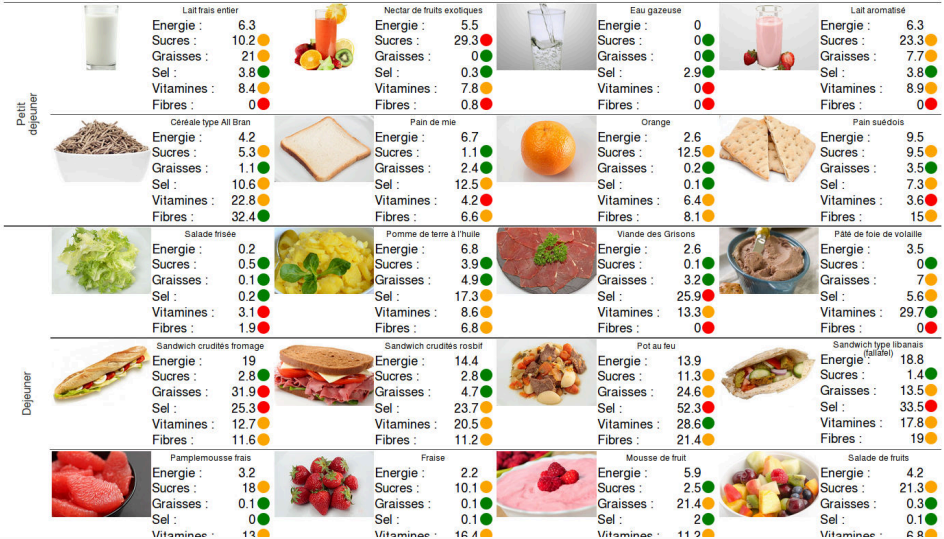
Colors



Numbers + colors



A number + colors screen, 7 constraints



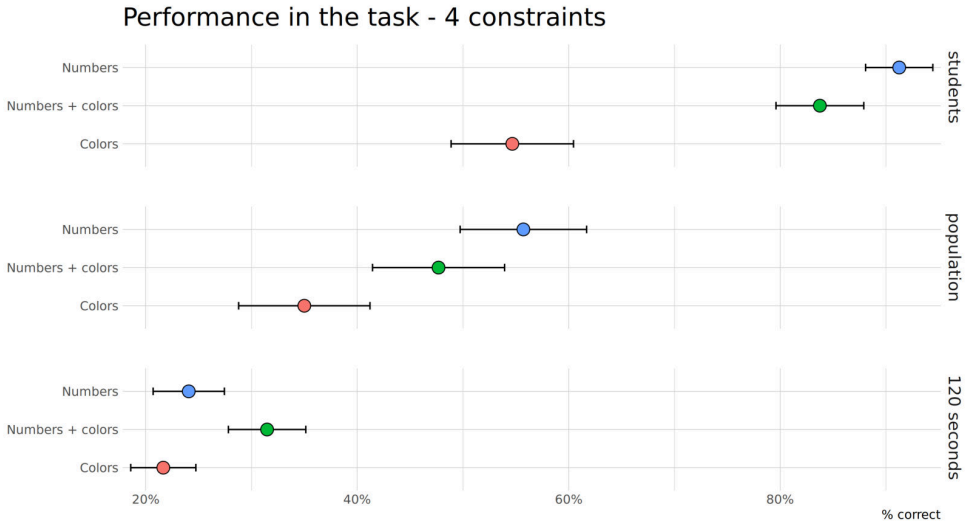
Two populations, three conditions

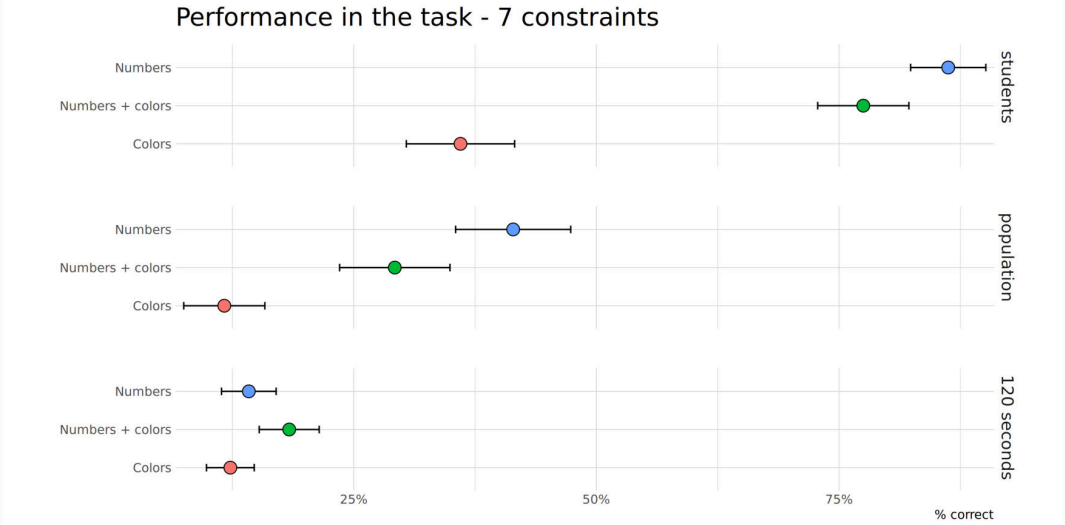
To investigate the role of

- **cognitive resources** and
- **time**

we run three conditions:

Students	Highly skilled engineering students, no time limit, paper and pencil
Population	Population at large, no time limit, paper and pencil
120 seconds	Population at large, 120 seconds, NO paper and pencil


Performance in the task - 4 constraints
Numbers
Numbers + colors
Colors
students
Numbers
Numbers + colors
Colors
population
Numbers
Numbers + colors
Colors
120 seconds
20%
40%
60%
80%
% correct


Performance in the task - 7 constraints
Numbers
Numbers + colors
Colors
students
Numbers
Numbers + colors
Colors
population
Numbers
Numbers + colors
Colors
120 seconds
25%
50%
75%
% correct

What do we learn?

If time is **unlimited**:

- Numbers win
- Especially so for highly skilled
- But also for general population

If time is **limited**:

- Numbers and colors equal
- Number + colors overall better
- Dismal performance in all cases

Study 2:*fast* vs. *slow* decisions [WIP – L. Muller]



Food choice is both fast & slow

Food choice : fast

Health goals : slow

Labels are both fast & slow

numbers : slow

analytic : slow

colors : fast

aggregate : fast

Nutrition Facts

Serving Size 2 CUPS (30g)
Servings per Container VARIED

Amount per Serving	
Calories 150	Calories from Fat 70
	% Daily Value*
Total Fat 7g	**11%**
Saturated Fat 1.5g	**6%**
Cholesterol 0mg	**0%**
Sodium 120mg	**5%**
Total Carbohydrate 20g	**7%**
Dietary Fiber 4g	**15%**
Sugars 9g	
Protein 1g	

Vitamin A 0% • Vitamin C 0%

Calcium 0% • Iron 2%

* Percent Daily Values are based on a 2,000 calorie diet. Your daily values may be higher or lower depending on your calorie needs:

	Calories	2,000	2,500
Total Fat	Less than	65g	80g
Sat Fat	Less than	20g	25g
Cholesterol	Less than	300mg	300mg
Sodium	Less than	2,400mg	2,400mg
Total Carbohydrate		300g	375g
Dietary Fiber		25g	30g

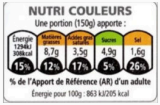
Calories per gram:
Fat 9 • Carbohydrate 4 • Protein 4

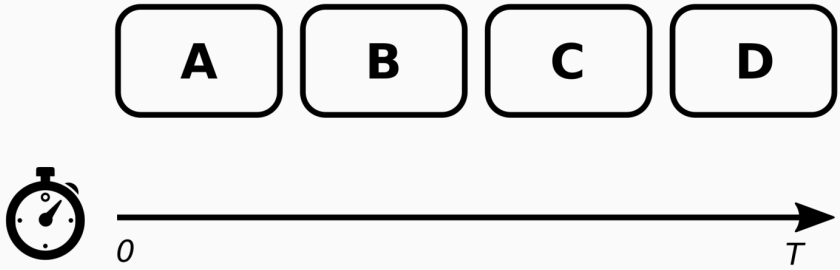


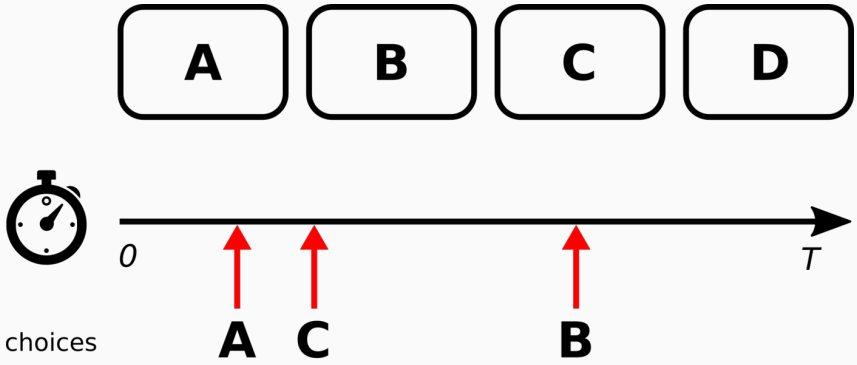


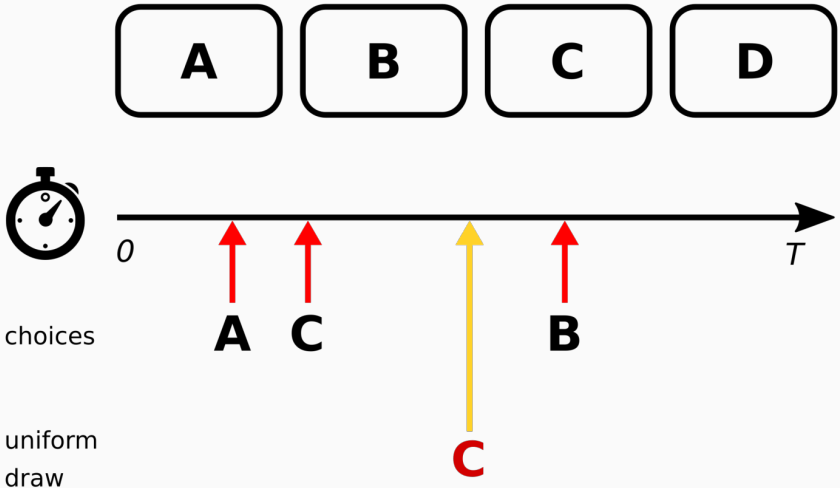


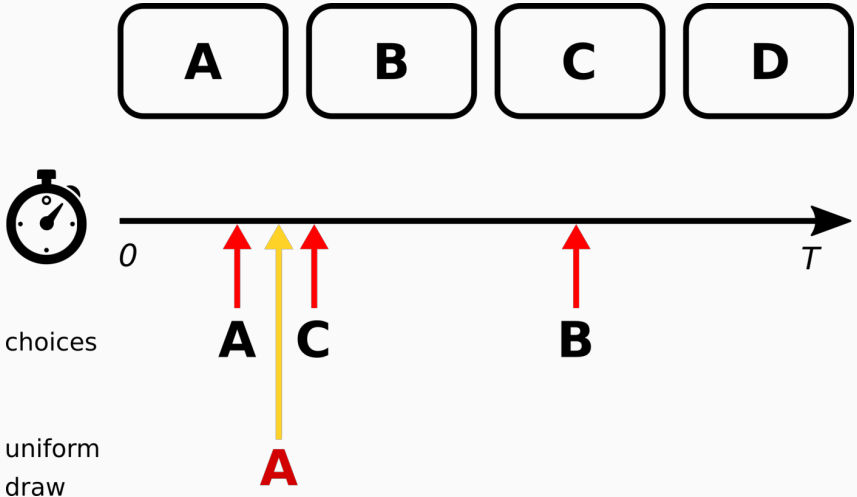


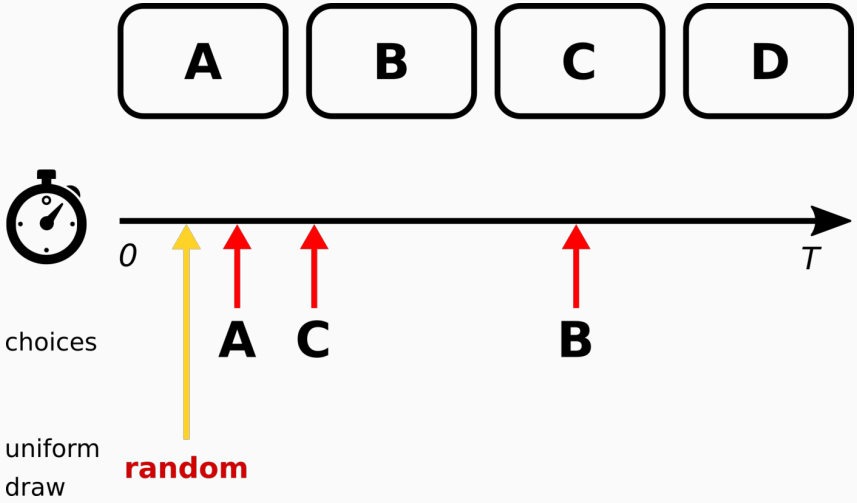



A
B
C
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A
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C
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choices
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C
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choices
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uniform
draw
C


A
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choices
A
C
B
uniform
draw
A


A
B
C
D
0
T
choices
A
C
B
uniform
draw
random

Our design

- No choice $\Rightarrow$ random choice: incentive to *fast* reply
- Time is ticking: incentive to *change* first decision upon reflection

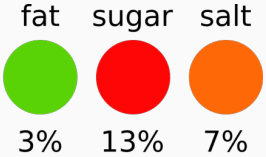
- fast to slow *endogenous* (usually: exogenous)
- Data reveal choice *process* (usually: outcome)

This allows us to:

- tell apart how different labels tap on different heuristics
- measure how much *faster* colors are
- assess if numbers do a better job, and when

Labels







fat sugar salt

3% 13% 7%

Barres au chocolat au lait et aux céréales



Choisir

Barres chocolatées fourrées au lait et aux noisettes



Choisir

Barres chocolatées au caramel

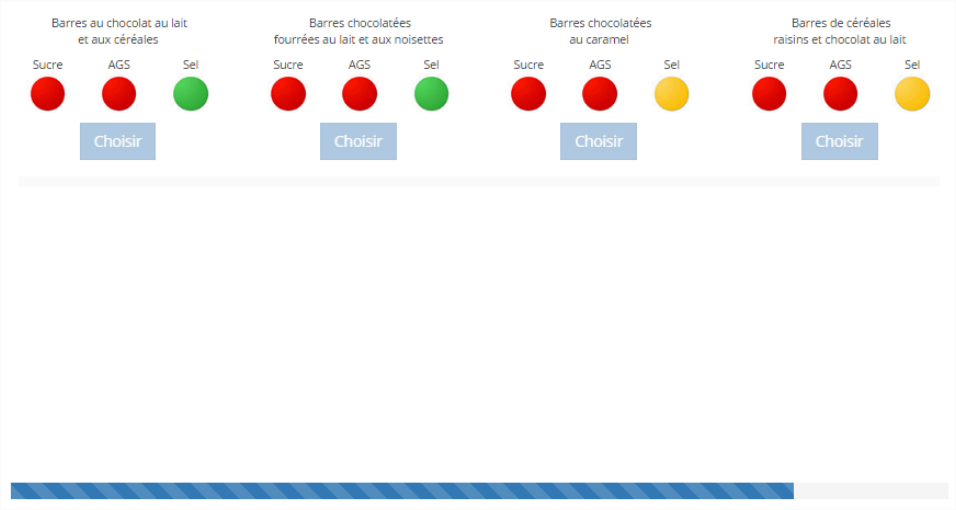


Choisir

Barres de céréales raisins et chocolat au lait



Choisir


Barres au chocolat au lait
et aux céréales
Sucre
AGS
Sel
Choisir
Barres chocolatées
fourrées au lait et aux noisettes
Sucre
AGS
Sel
Choisir
Barres chocolatées
au caramel
Sucre
AGS
Sel
Choisir
Barres de céréales
raisins et chocolat au lait
Sucre
AGS
Sel
Choisir

Barres au chocolat au lait et aux céréales

Sucres	AGS	Sel
55%	110%	5%

Choisir

Barres chocolatées fourrées au lait et aux noisettes

Sucres	AGS	Sel
46%	87%	5%

Choisir

Barres chocolatées au caramel

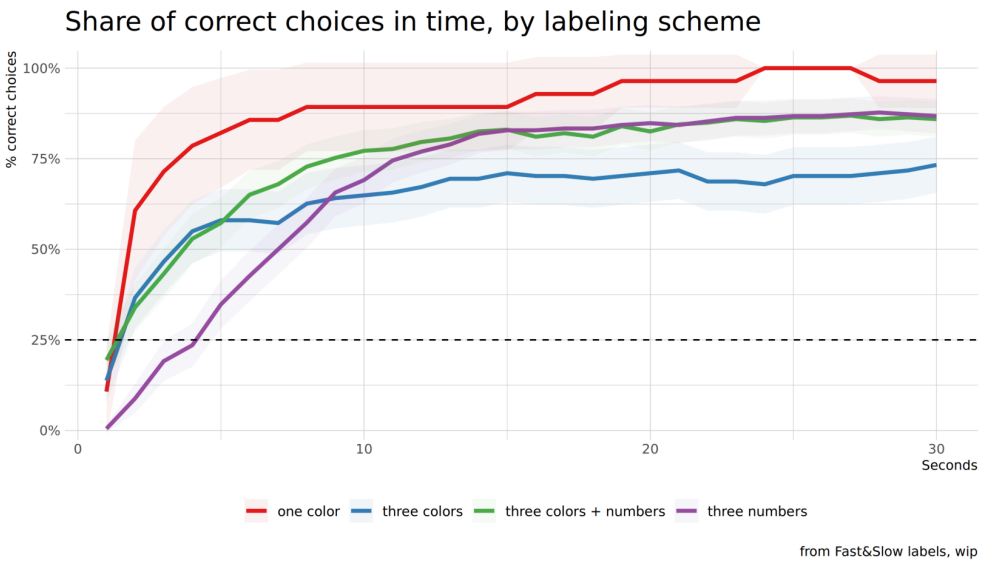
Sucres	AGS	Sel
69%	42%	7%

Choisir

Barres de céréales raisins et chocolat au lait

Sucres	AGS	Sel
34%	27%	9%

Choisir

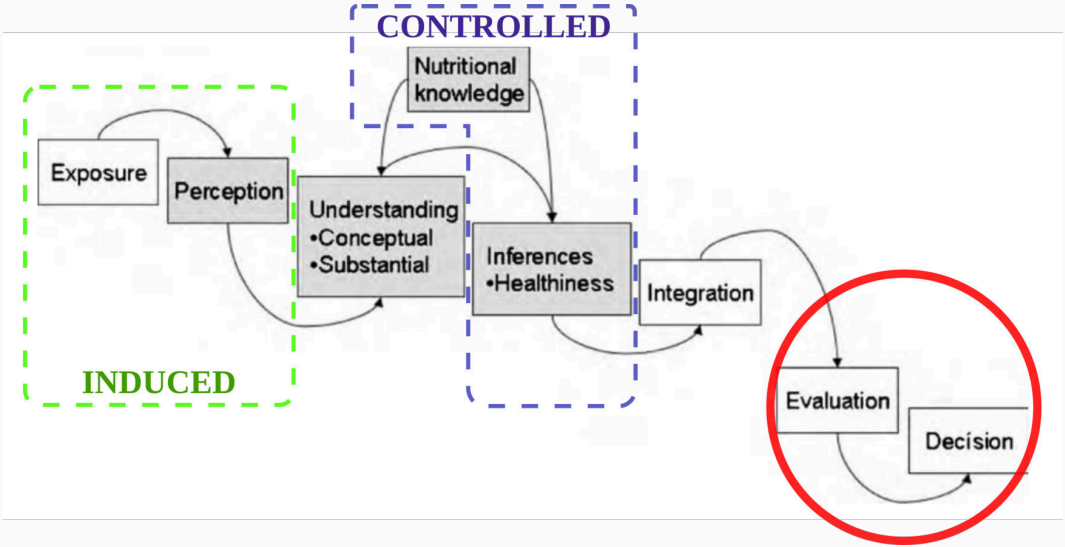

Share of correct choices in time, by labeling scheme
% correct choices
100%
75%
50%
25%
0%
0
10
20
30
Seconds
one color
three colors
three colors + numbers
three numbers
from Fast&Slow labels, wip

What do we learn?

- Trade off time/accuracy
- Heuristics give way to computation in time
- Indirect evidence of different cognitive processes
- We explicitly measure 'how more intuitive' colors are

Part 2: building counterfactuals

Building counterfactuals



The usual design(s)

Preferences are back – and they are **key**:

- WTP studies
- Auctions
- Choice experiments

Study 3: shopping with labels [ERAE 2019 – L. Muller, B. Ruffieux, A. Lacroix]

Some context



NUTRI-SCORE

A B C D E

Inserm

La science pour la santé
From science to health



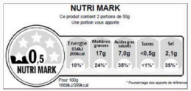
ania

Association Nationale des Industries Alimentaires

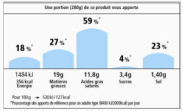
fcd Fédération du Commerce et de la Distribution

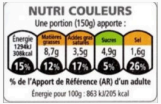
The question











French Ministry of Health – 2016

- Which FoPL to choose?
- How large is the effect?

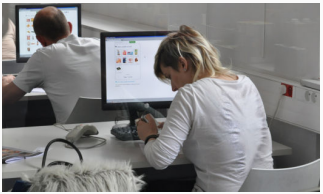
- A RCT in 60 French supermarket
- A large lab experiment (**us!**)

Our setup

Paper catalog



Computer interface



Real products



- Subjects shop for real in the lab
- For two days for their household

- $\sim \frac{1}{4}$ of product supply available
- chosen + we have it $\Rightarrow$ buy

A large and representative catalog

- 290 products
- 37 food categories
- custom e-shopping interface
- barcode scanners on the desk
- price, quantity, picture (label) up front
- nutritional table and ingredient list available upon clicking

1138 Chercher

Votre caddie actuel :

Aucun produit dans ce panier.

Terminer

4,25 €
260 g
16,35 €/Kg

1 Ajouter au caddie

2 Steaks hachés pur boeuf 15% mg



Ingrédients Valeurs nutritionnelles

1814 Chercher

1814

Votre caddie actuel :


x 1 = 4.25 €


x 1 = 3.95 €


x 1 = 1.17 €


x 1 = 2.83 €


x 3 = 1.83 €

5 Articles

Total = 14.03 €

Terminer

1,62 €
180 g
9,00 €/Kg

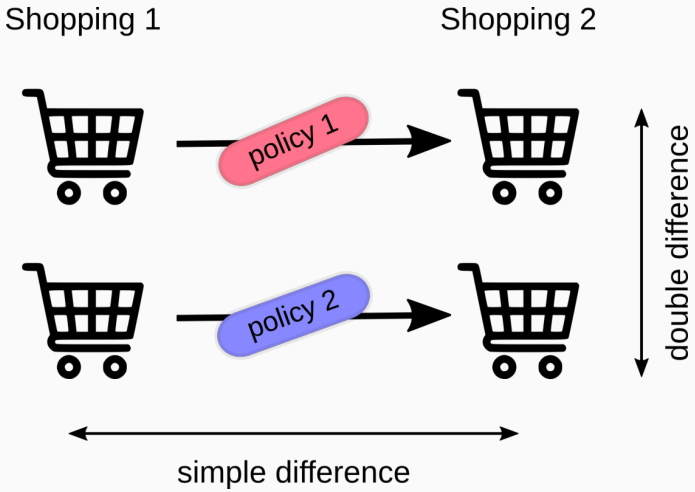
1 Ajouter au caddie

Frites surgelées pour micro-ondes



Ingrédients | Valeurs nutritionnelles

Experimental design: difference-in-difference



Metrics: nutrition

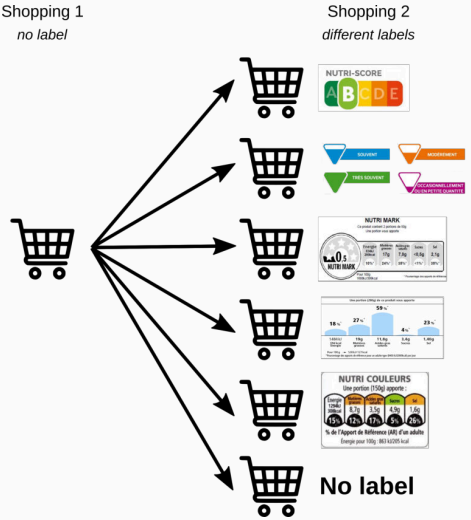
We use the **scoreFSA** normalized by **caloric content**.

For each shopping cart i, for each subject j, for each product p:

$$scoreFSA_{ij} = \frac{\sum_p Kcal_{pij} \cdot FSA_{pij}}{\sum_p Kcal_{pij}},$$

We focus on Δ**FSA**, the *difference* between carts 1 and 2.

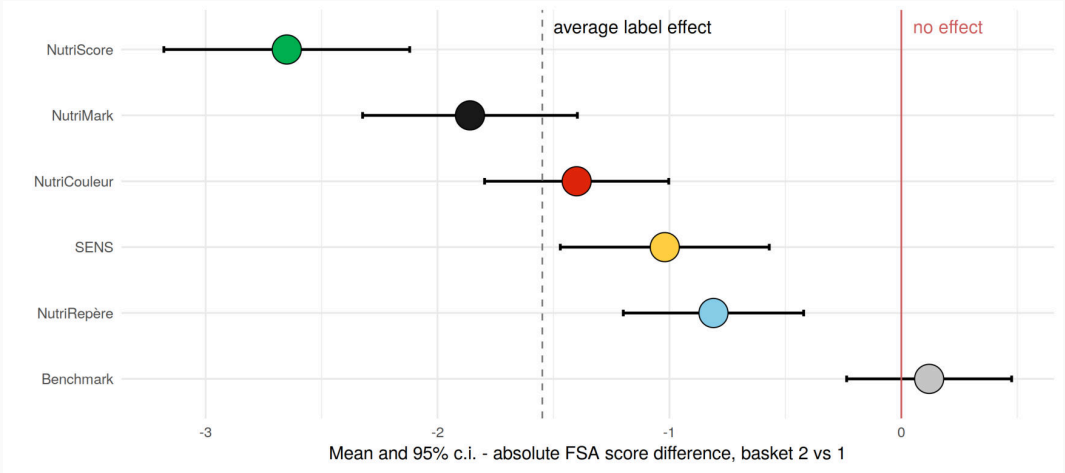
Treatments


Shopping 1
no label
Shopping 2
different labels
NUTRI-SCORE
A B C D E
SOUVENT
MODÉREMENT
TRÈS SOUVENT
OCCASIONNELLEMENT OU EN PETITE QUANTITÉ
NUTRI MARK
NUTRI COULEURS
Une portion (150g) apporte :
% de l'Apport de Référence (AR) d'un adulte
No label

A large and representative subject pool

- 691 subjects
- $\sim$ 110 for each of 6 treatments
- sample issued from the general population
- (recruiting agency boosted our reach into all socio-economic statuses)
- roughly representative

NutriScore leads, by far



But there is heterogeneity



What do we learn?

- Color-coded, summary labels perform best
- (but only if they directly relate to quality – not SENS)
- Number-based, analytic labels perform worse

Policy implications

- NutriScore officially selected as French (and Spanish) standard
- Adopted by Auchan, Fleury Michon, Leclerc, Casino, Nestlé
- The very idea of FOP labels validated
- NutriScore is being proposed by France as EU standard

Study 3: Lab vs. Field
[WIP – poster by L. Muller]

The two studies at a glance

	Lab	Field
Location	Grenoble	Paris couronne, Nord, Lyon
Supermarkets	"1"	60
Task	shop for two days' worth	shop
Real purchases	"yes"	yes
Measure	FSA score for 2000Kcal	FSA score for 2000Kcal
Design	Diff-in-diff	Diff-in-diff
Time frame	2x, same session	5 weeks, 1 year apart
Participants	691	171.827
Products (of which labeled)	290 (all)	3586 (1266)
Food categories	37	4
Purchases	27.882	1.668.301
Manpower needed	8	$\sim$ 100
Cost	$\sim$100k	$\sim$4 million

Field study: "instructions"



Field study: product display





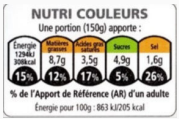


Field study: alerting the subject to the experimentation





Results: the lab as a magnifying glass

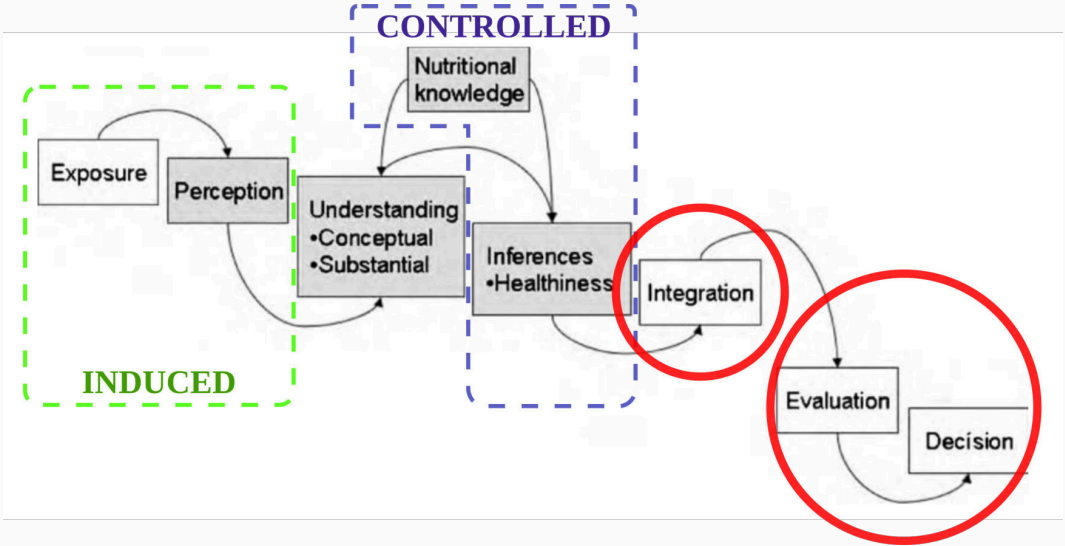
Label	△score FSA		Corr	Zoom
	Field	Lab		
	-0.142*	-2.766***		19x
	-0.115	-1.513*	0.88	13x
	-0.062	-1.140		18x
	-0.024	-0.924		38x

What is going on? Which one is the *correct* estimate?

Demand effect	similar in both experiments
Strategic behavior	
Social desirability bias	
Game form misconception & complexity	mostly same simple everyday task
Incentive compatibility	same in both experiments
Subject pool differences	not really
Self-selection	not much, but our lab sample *is* selected
Focality and attention	**stark** difference
Time contraction	**stark** difference and generates focality

Study 4: labels or prices?
[WIP – L. Muller, B. Ruffieux]

Integrating different policies



A bit of context

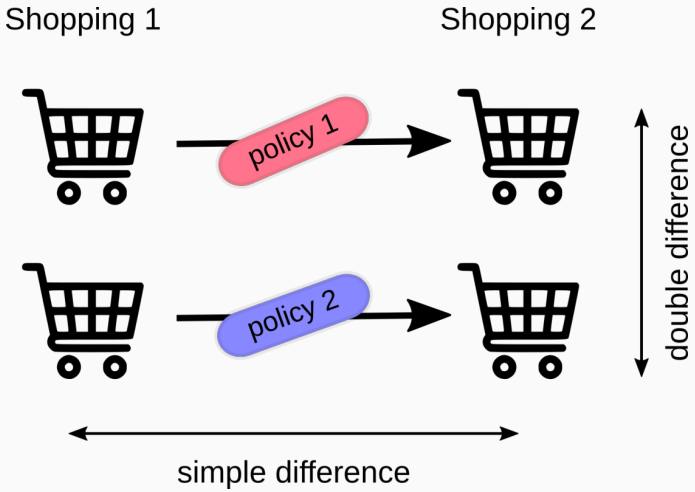
- World Bank: strongly tax unhealthy foods (Shekar and Popkin 2020)
- WHO: introduce dietary taxes on unhealthy food of minimum 20%
- India and Mexico tax unhealthy food & beverages (India : tax of 28%).

Question and design

- Suppose we want to couple a **label** with an **incentive** scheme
- e.g. tax unhealthy (soda tax) and subsidize healthy food.
- Does it work? How?
- Will the intervention be (sub/super)additive?
- i.e. label **or** price $\gtrless$ label **plus** price?

Exact same design as Study 2

Experimental design: difference-in-difference



Metrics: expenditure

We use the **expenditure** on a basket, normalized by **2000Kcal**.

For each shopping cart i, for each subject j, for each product p:

$$expenditure_{ij} = 2000 * \frac{\sum_p Price_{pij}}{\sum_p Kcal_{pij}},$$

We focus on Δ**Expenditure**, the *difference* between carts 1 and 2.

Metrics: state investment

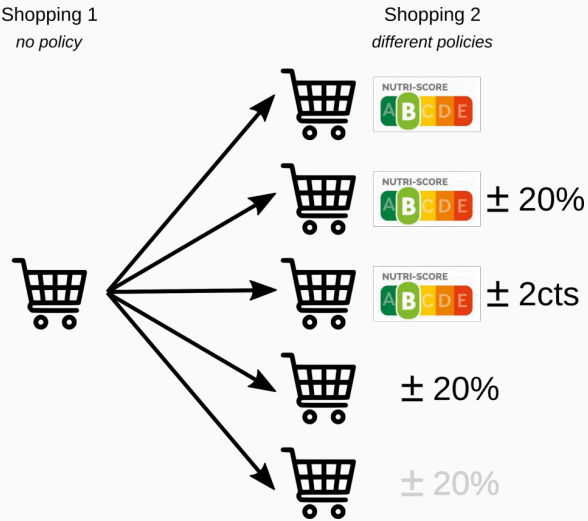
We use the total amount of subsidies minus the total amount of tax revenue, per consumer.

Since consumers had to buy for 2 days, we divide by 2 to get a daily cost.

for each subject j, for each product p:

$$stateaid_j = \frac{\sum_p (tax_{pj} - subsidy_{pj})}{2},$$

Treatments


Shopping 1
no policy
Shopping 2
different policies
NUTRI-SCORE
A B C D E
± 20%
NUTRI-SCORE
A B C D E
± 2cts
NUTRI-SCORE
A B C D E
± 20%
± 20%

A *large* price change: ±10% or 20%



A *small* price change: ± 1 or 2cents



Stimuli

Référence
Sans Nutri-Score
sans Bonus-Malus



Traitement 1
Nutri-Score
sans Bonus-Malus



Traitement 2
Nutri-Score
avec Bonus-Malus explicite
ad valorem
de niveau élevé



Traitement 3
Nutri-Score
avec Bonus-Malus explicite
par unité de
niveau symbolique



Traitement 4

Bonus-Malus implicite
ad valorem
de niveau élevé



Traitement 5

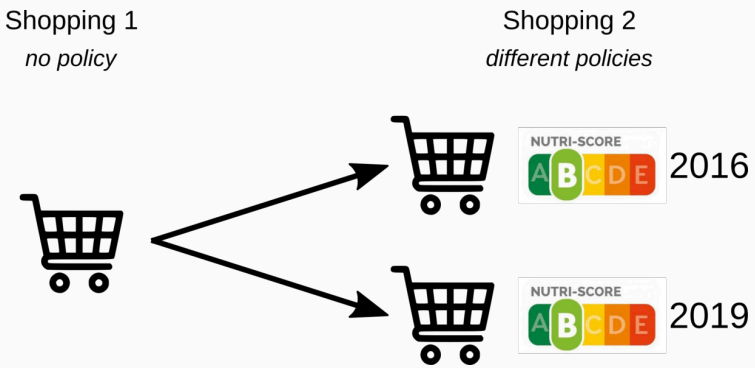
Bonus-Malus explicite
ad valorem
de niveau élevé



A large(ish) and representative subject pool

- 386 subjects
- $\sim$ 75 for each of the 5 treatments
- sample issued from the general population
- roughly representative (++women, ++educated)

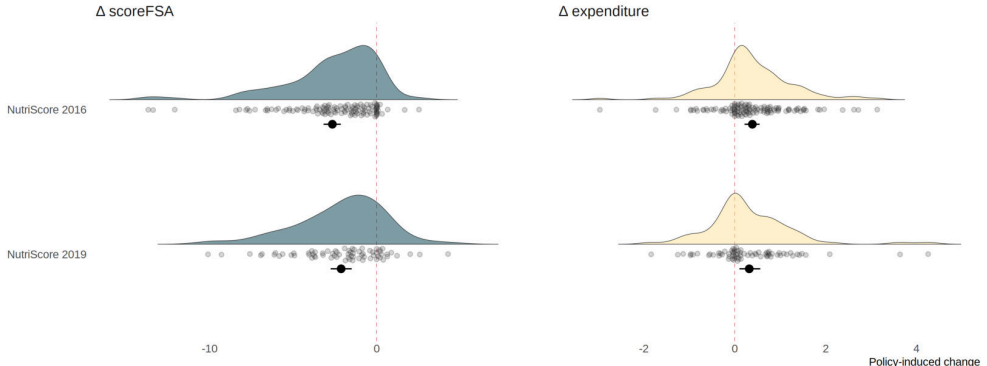
Replication



Hp: we will *replicate* the 2016 findings re. NutriScore

Results: replication

	ScoreFSA				Expediture			
	cart 1	cart 2	difference	p-value	cart 1	cart 2	difference	p-value
NutriScore 2019	3.1 (2.95)	0.97 (3.07)	-2.13 (2.67)	0.205	5.85 (1.64)	6.17 (1.97)	0.32 (0.95)	0.621
NutriScore 2016	4.74 (3.43)	2.09 (3.47)	-2.65 (2.84)		5.24 (1.6)	5.63 (1.65)	0.39 (0.86)	



Policy mix additivity

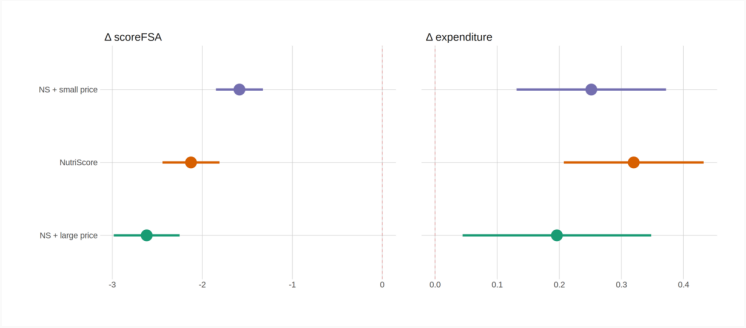


Hp: policies are subadditive: $A \mid B \leq f(A,B) \leq A + B$

Additivity: results

comparison	expenditure	scoreFSA
large vs small	0.394	0.020
NS vs large	0.243	0.411
NS vs small	0.851	0.171

Wilcoxon rank-sum p-values



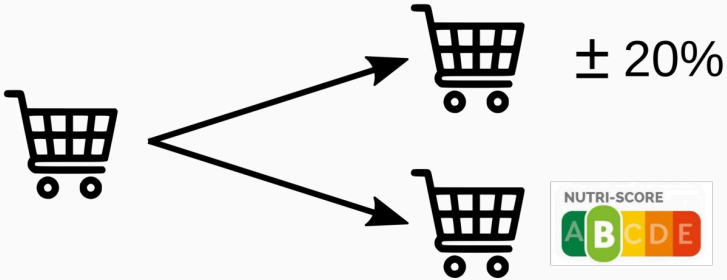
Information *vs.* incentives

Shopping 1

no policy

Shopping 2

different policies

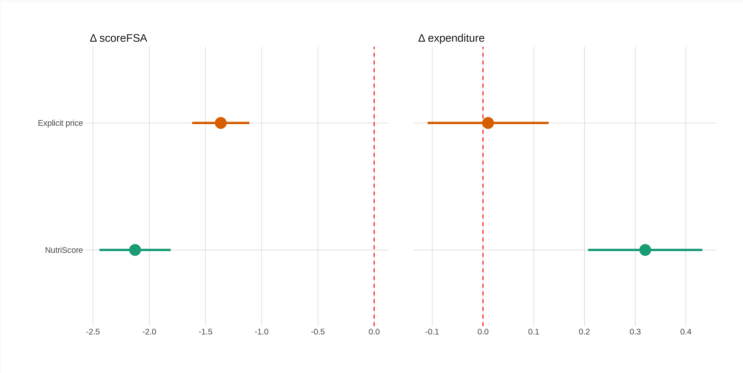


Hp: Labels have a higher impact than prices

Information *vs.* incentives: results

indicator	p.value
expenditure	0.005
scoreFSA	0.099

Wilcoxon rank-sum p-values



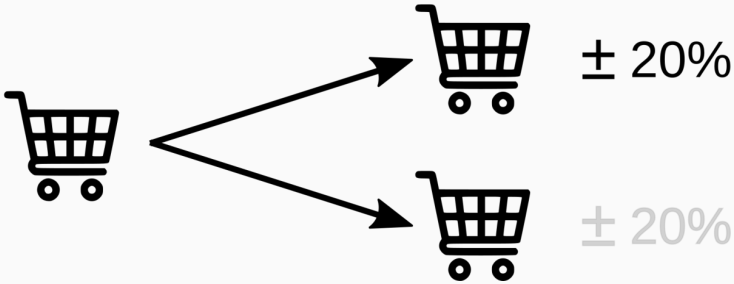
Price salience

Shopping 1

no policy

Shopping 2

different policies

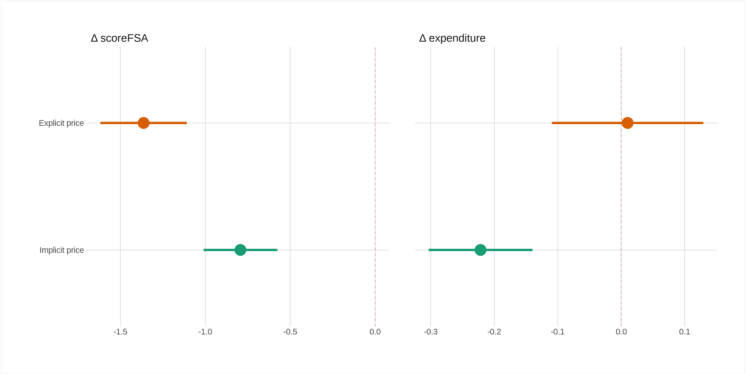


Hp: *salient* price changes have a larger impact

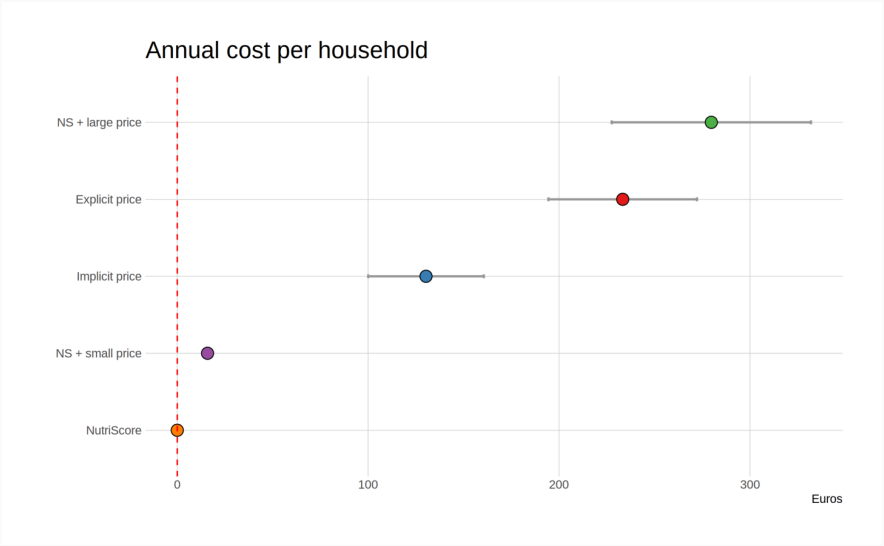
Price salience: results

indicator	p.value
expenditure	0.564
scoreFSA	0.046

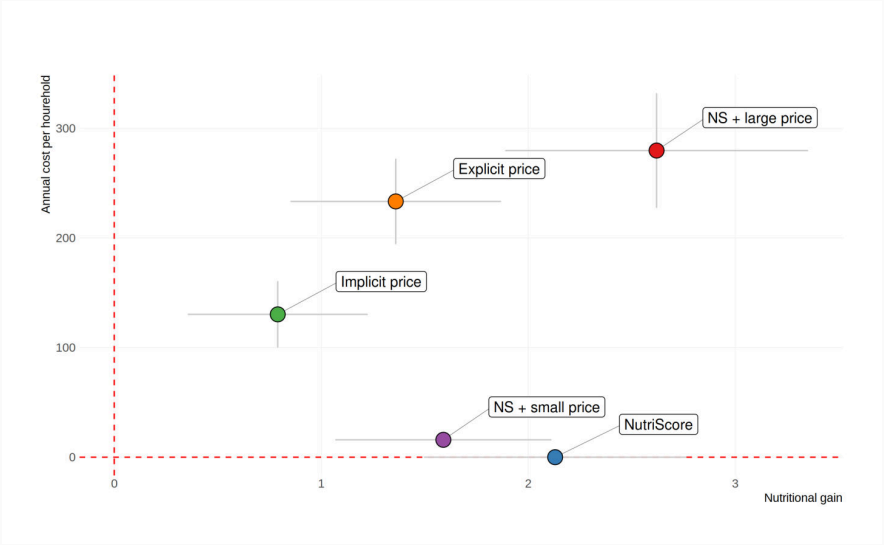
Wilcoxon rank-sum p-values



Welfare analysis: cost for the state



Welfare analysis: trade-offs



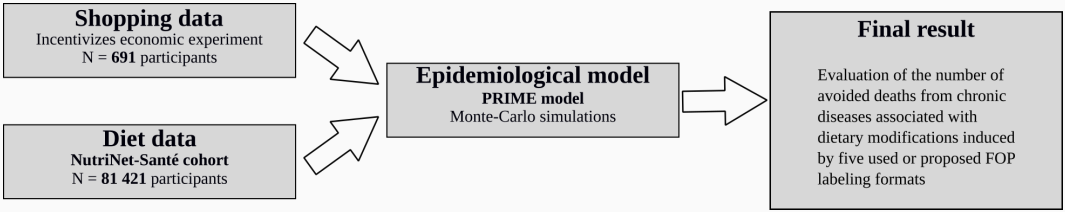
What do we learn?

- Nutritional policies are subadditive
- *Too small* an incentive reduces the effect (Gneezy & Rustichini)
- Price policies have better be explicit (Chetty et al.)
- Labeling appears as more cost-effective than the policy mix
- ...still, it's just the lab!

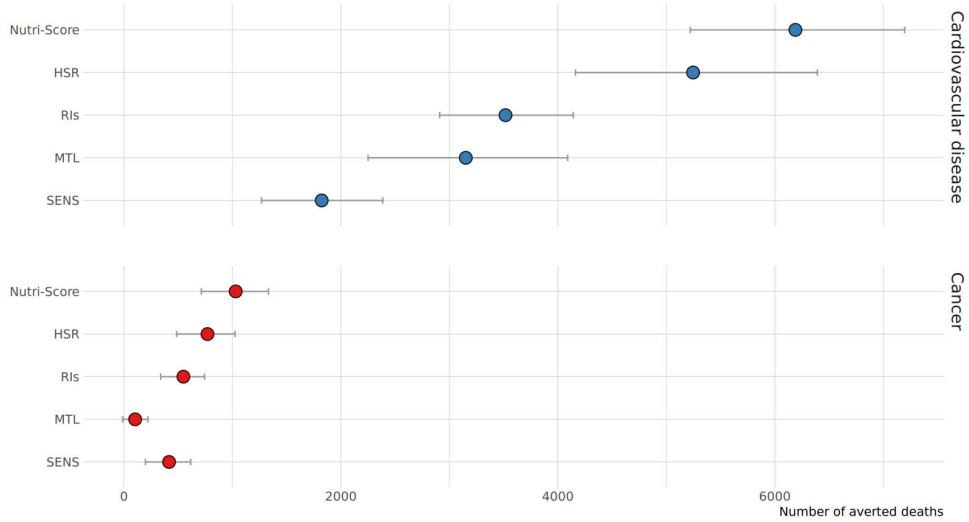
Part 4: does it matter?

Study 5: epidemiology
[IJBNPA 2019 – S. Hercberg et al]

From micro to macro



Estimated number of averted deaths, France



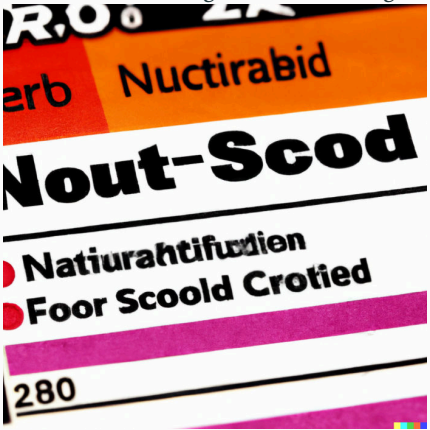
What do we learn?

- Labels have non-negligible impacts on mortality
- Results from the lab can be used to feed macro models
- Better, intuitive labels are used and save lives.

Part 5: the future

Asking an AI

"Nutritional labeling for human beings"



"colorful impactful relevant simple effective nutritional labeling"



Some open questions: consumers

How externally valid are our results?

- Andrew Gelman: if all these biases were true, they would dwarf main effects
- How to integrate all these labels?
- Label proliferation
- Information overload

- Cultural arena: the battle for label perception
- Nutrition vs tradition
- A contrarian view from Italy

Some open questions: firms

There is a world beyond consumers: firms

- Price discrimination
- Multiple labeling
- Labels as anti-competitive devices

- Labels working for the *wrong* reasons
- Normative messages
- "*Bisogna che tutto cambi, affinché tutto resti uguale*"


thank
you